

Control work 6, and work on the mistakes

Parallel lines, the angle sum and right-angled triangles in one paper.

LESSON 96–97

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 6

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Name a pair of angles at a transversal before quoting any rule about it.
- Use the property when parallelism is given and the criterion when it is wanted.
- Find angles of triangles, including from an exterior angle.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 226–281
Cambridge	Stage 8, pp. 44–56
Workbook	Control paper M6

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Name the three pairs of angles at a transversal and say what each does when the lines are parallel	3	L82–85
2	$a \parallel b$ and one angle is 112° : find all eight angles	4	L84–85
3	Co-interior angles are $4x$ and $5x$: find both	4	L84–85
4	The angles of a triangle are in the ratio $2 : 3 : 4$: find them	4	L86–87
5	An exterior angle is 125° and one remote interior angle is 47° : find the other two interior angles	5	L86–87
6	Hypotenuse 16 cm , one angle 30° : find the opposite leg and the third angle	5	L88–89

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the word “co-interior” alone; Q2 one for noticing that only two sizes appear; Q5 two for using the *remote* pair and one for the check against 180° ; Q6 one for the 30° rule being quoted by name.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
pair misnamed	“alternate” for two angles on the same side	check the position first
co-interior treated as equal	$4x = 5x$	$4x + 5x = 180^\circ$
criterion used as a property	“alternate, so parallel, so equal”	parallelism was given — use the property
all three angles added to the exterior angle	$125^\circ = 47^\circ + b + c$	the two <i>remote</i> angles only
acute angles subtracted from 180°	$180^\circ - 30^\circ = 150^\circ$	$90^\circ - 30^\circ = 60^\circ$
30° rule inverted	the leg is twice the hypotenuse	it is half
no check	three angles that do not total 180°	add them back

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

BLOCK	THE SENTENCE
parallel lines	no common point, and exactly one parallel through an outside point
the criteria	an angle fact proves parallelism
the properties	parallelism proves an angle fact
the angle sum	180°
the exterior angle	equal to the two remote interior angles
right-angled triangles	the acute angles add to 90°
the 30° rule	that leg is half the hypotenuse

LOOKING FORWARD

The rest of Quarter III is algebra: the abridged multiplication formulae, algebraic fractions and linear equations. Geometry returns in Quarter IV with construction problems.

02 Worked examples

EXAMPLE 1 Model answer, Q3: co-interior angles $4x$ and $5x$.

- 1 Co-interior angles on parallel lines add to 180° .
- 2 $4x + 5x = 180^\circ$
- 3 $9x = 180^\circ$, so $x = 20^\circ$.
- 4 The angles are 80° and 100° .

ANSWER

80° and 100°

EXAMPLE 2 Model answer, Q5: exterior 125° , one remote interior angle 47° .

- 1 The exterior angle equals the sum of the two remote angles.
- 2 $125^\circ - 47^\circ = 78^\circ$
The second remote angle.
- 3 The adjacent interior angle is $180^\circ - 125^\circ = 55^\circ$.
- 4 Check: $47^\circ + 78^\circ + 55^\circ = 180^\circ$ ✓

ANSWER

78° and 55°

EXAMPLE 3 Model answer, Q6: hypotenuse 16 cm , one angle 30° .

1 The leg opposite 30° is half the hypotenuse.

2 $16 \div 2 = 8\text{ cm}$

3 The acute angles add to 90° .

4 $90^\circ - 30^\circ = 60^\circ$

ANSWER

8 cm and 60°

03 Interactive model

Return Q1 first and have the class read the three names aloud; most of the lost marks in Q2 and Q3 turn out to be naming errors carried forward.

The chapter in twelve questions

Parallel lines have how many common points?

Parallels through a point off a line:

Equal alternate angles prove:

On parallel lines, co-interior angles:

are equal

add to 90°

add to 180°

add to 360°

With one angle of 112° , the eight angles take:

one size

two sizes

four sizes

eight sizes

Co-interior $4x$ and $5x$ give x equal to:

10

20

36

45

The angles of a triangle add to:

90°

180°

270°

360°

Ratio 2 : 3 : 4 gives:

$40^\circ, 60^\circ, 80^\circ$

$30^\circ, 60^\circ, 90^\circ$

$45^\circ, 60^\circ, 75^\circ$

$20^\circ, 30^\circ, 40^\circ$

An exterior angle equals:

the adjacent angle

the two remote angles

all three

90°

The acute angles of a right triangle add to:

45°

90°

180°

270°

Hypotenuse 16, angle 30° : the opposite leg is:

4

8

16

32

The longest side of any triangle faces:

the smallest angle

the largest angle

the right angle only

any angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Transversal	Kesuvchi	Секущая
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Angle sum	Burchaklar yig'indisi	Сумма углов
Exterior angle	Tashqi burchak	Внешний угол
Hypotenuse	Gipotenuza	Гипотенуза
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 asks first for:

a calculation

the names of the pairs

a construction

a proof

In Q2 the eight angles take:

one size

two sizes

four sizes

eight sizes

Q3 uses:

equal co-interior angles

co-interior angles adding to 180°

the angle sum

the exterior angle

Q4 sets up:

$$9x = 180$$

$$9x = 360$$

$$x = 180$$

$$3x = 180$$

Q5 uses:

all three interior angles

the two remote angles

the adjacent angle only

the angle sum alone

Q6 quotes:

the angle sum

the 30° rule

the criterion

the triangle inequality

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. On parallel lines, alternate angles are

ANSWER equal

2. On parallel lines, co-interior angles add to

ANSWER 180°

3. One angle is 112° : its supplement

ANSWER 68°

4. Angles of a triangle add to

ANSWER 180°

5. Acute angles of a right triangle add to

ANSWER 90°

6. Hypotenuse 16, angle 30° : the opposite leg

ANSWER 8

7. Exterior 125° : the adjacent interior angle

ANSWER 55°

Medium · the standard question

7 PROBLEMS

1. Co-interior $4x$ and $5x$

ANSWER 80° and 100°

2. Ratio $2 : 3 : 4$

ANSWER $40^\circ, 60^\circ, 80^\circ$

3. Exterior 125° , one remote angle 47° : the other

ANSWER 78°

4. $a \parallel b$ and one angle 112° : the alternate angle

ANSWER 112°

5. $a \parallel b$ and one angle 112° : the co-interior angle

ANSWER 68°

6. Right triangle with an angle of 58° : the third angle

ANSWER 32°

7. Sides 4, 6, 11: a triangle?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. Alternate angles $6x - 5$ and $4x + 25$ on parallel lines

ANSWER $x = 15$

2. An exterior angle is 4 times its adjacent interior angle

ANSWER 144° and 36°

3. A right triangle has acute angles in the ratio 2 : 7

ANSWER 20° and 70°

4. Co-interior $2x + 30$ and $x + 60$

ANSWER $x = 30$

5. One angle of a triangle is the sum of the other two: what kind of triangle is it?

ANSWER Right-angled

6. A transversal cuts two parallel lines at 90° : all eight angles are

ANSWER 90°

7. Angles x , $x + 20$, $x + 40$

ANSWER 40° , 60° , 80°

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Draw a transversal across two parallel lines, mark 118° and label all eight angles.
3. Co-interior angles are $7x$ and $2x$. Find both.
4. An exterior angle is 132° and one remote interior angle is 61° . Find the other two.
5. Write one sentence for each of the seven blocks on the chapter map.